

Geometric Series Worksheet

Twenty finite, infinite, shifted-index, decimal, modeling, and error-analysis problems.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/worksheets/geometric-series/>

Name: _____ **Date:** _____

1. Find the common ratio of 3, 12, 48, ...

2. Find a_8 when $a_1=5$ and $r=2$.

3. Evaluate $\sum_{k=0}^5 3(2)^k$.

4. Evaluate $\sum_{k=0}^4 16(1/2)^k$.

5. Evaluate $\sum_{k=0}^{\infty} 12(1/3)^k$.

6. Evaluate $\sum_{k=0}^{\infty} 8(-1/2)^k$.

7. Decide whether $\sum_{k=0}^{\infty} 5(1.02)^k$ converges.

8. Evaluate $\sum_{n=3}^{\infty} 2(1/4)^n$.
9. Rewrite $\sum_{n=1}^{\infty} 3^{1-n}$ in ar^k form and sum it.
10. Write 0.333... as a fraction.
11. Write 0.272727... as a fraction.
12. Write 0.145145... as a fraction.
13. A ball rebounds 70% of each previous height after a 10 m drop. Find total vertical distance.
14. Deposit 100 dollars at the end of each year for four years at 5 percent. Find the value immediately after the fourth deposit.
15. A square has area 1; each stage shades one fourth of the remaining area. Find total shaded area.

16. Find S_n for $7+7(0.8)+7(0.8)^2+\dots$
17. An infinite geometric series has first term 6 and sum 15. Find r .
18. An infinite geometric series has ratio $-1/4$ and sum 8. Find its first term.
19. A student uses $a/(1-r)$ for $r=2$. Diagnose the error.
20. Evaluate $\sum_{n=2}^6 5(3)^{n-2}$.